

EXPLORATORY DATA ANALYSIS

Project 2 – Industrial Training Kit

E-Commerce Order Dataset (2023–2025)

Senior Data Analyst Report
DecodeLabs | Batch 2026
Prepared following Forensic EDA Framework

Total Orders	Total Revenue	Avg Order Value	Median Order Value
1,200	\$1,264,762	\$1,053.97	\$823.62

1. Problem Statement

This EDA investigates a 1,200-row e-commerce order dataset spanning January 2023 to June 2025. The core question is: **What are the underlying patterns, distributions, trends, and anomalies in order behaviour that can inform revenue optimization, marketing allocation, and operational risk?**

Following the DecodeLabs Forensic Architecture of EDA, the analysis treats the dataset as a repository of digital evidence. We apply the IPO framework (Input → Process → Output) and translate statistical findings into business diagnoses using the “So What?” test.

2. Methodology & Data Forensics

2.1 Dataset Overview

- **Source:** DecodeLabs Project 2 Dataset (Excel)
- **Shape:** 1,200 orders × 14 columns
- **Time span:** 2023-01-01 → 2025-06-30
- **Unique customers:** 1,189 (near 1:1 order-to-customer ratio)
- **Key variables:** OrderID, Date, CustomerID, Product, Quantity, UnitPrice, PaymentMethod, OrderStatus, ItemsInCart, CouponCode, ReferralSource, TotalPrice

2.2 Data Quality Assessment

- **Missing values:** CouponCode has 309 nulls (25.75%). All other fields are complete.
- **Consistency check:** $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$ holds for all rows (max absolute difference < $1e-12$).
- **Duplicates:** None (OrderID is unique).
- **Data types:** Correctly inferred; Date is datetime64, numeric fields are float/int, categoricals are string.

2.3 Analytical Toolkit

Descriptive statistics (mean, median, quantiles), distribution shape (skewness, kurtosis), five-number summary, IQR and Z-score outlier detection, Pearson correlation, time-series aggregation, and categorical breakdowns. Visualizations generated with matplotlib/seaborn.

3. Key Findings

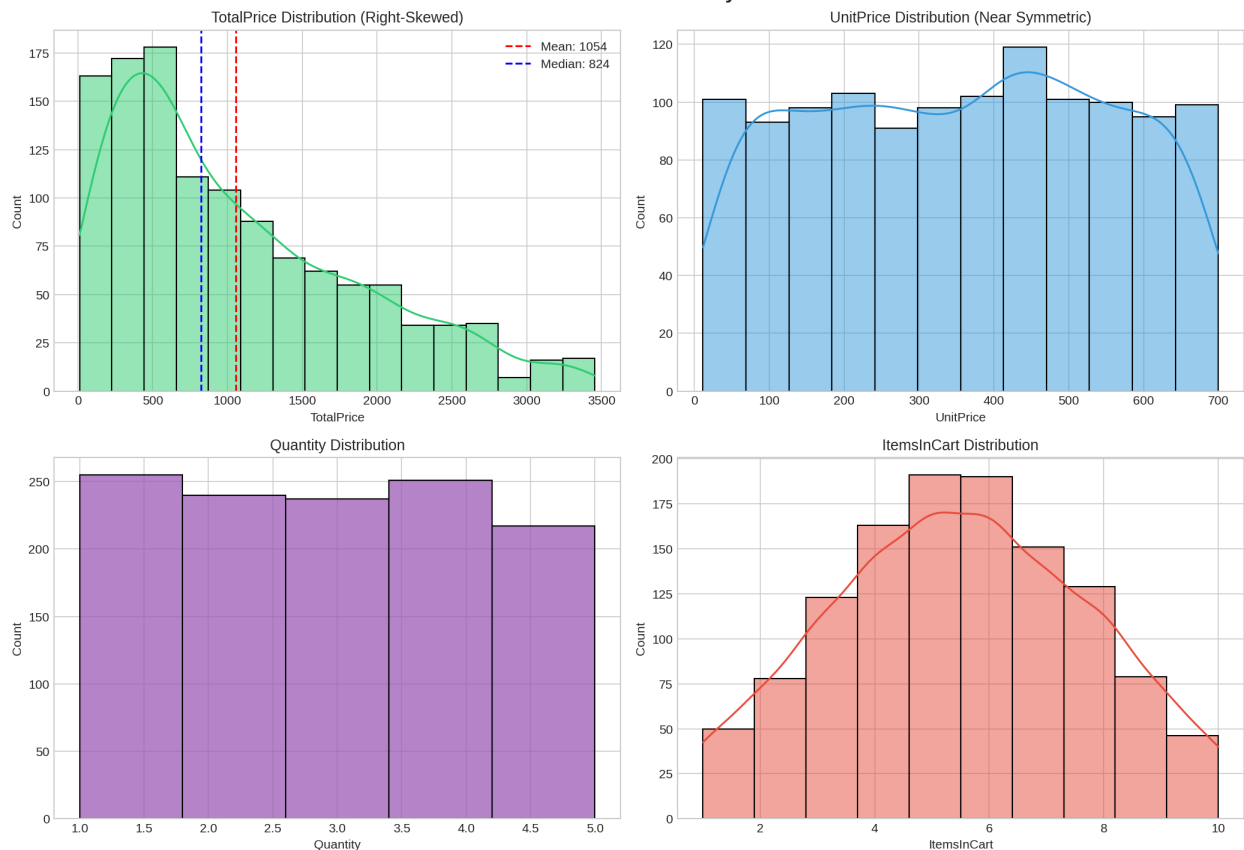
3.1 Basic Descriptive Statistics

Metric	Mean	Median	Std	Min	Max
Quantity	2.95	3.00	1.41	1	5
UnitPrice (\$)	356.41	364.21	197.18	11.39	699.93
ItemsInCart	5.49	5.00	2.28	1	10
TotalPrice (\$)	1,053.97	823.62	819.86	11.39	3,456.40

Interpretation: Mean TotalPrice (\$1,054) is substantially higher than the median (\$824), indicating a right-skewed distribution. This is the classic signature of a few high-value orders pulling the average upward. UnitPrice and Quantity are nearly symmetric.

3.2 Distribution Geometry

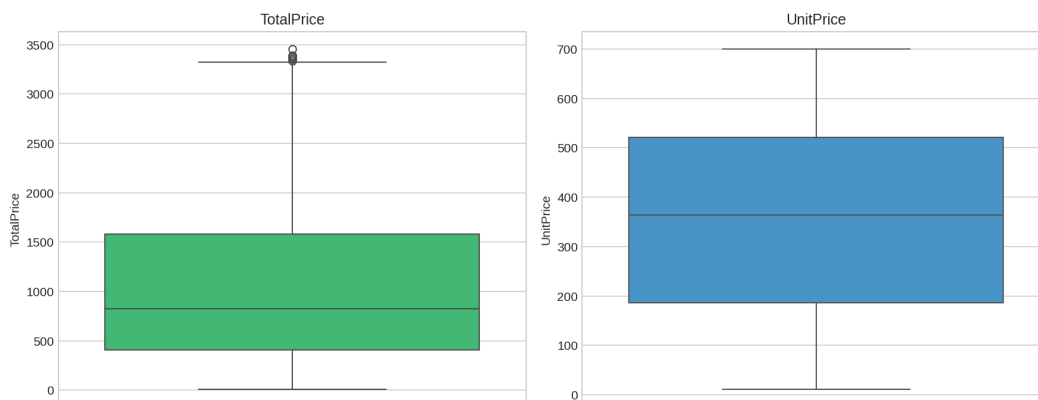
Univariate Distributions of Key Metrics



TotalPrice is right-skewed (skewness ≈ 0.89). Quantity and ItemsInCart are roughly uniform/symmetric within their discrete ranges. UnitPrice is nearly symmetric (skewness ≈ -0.03).

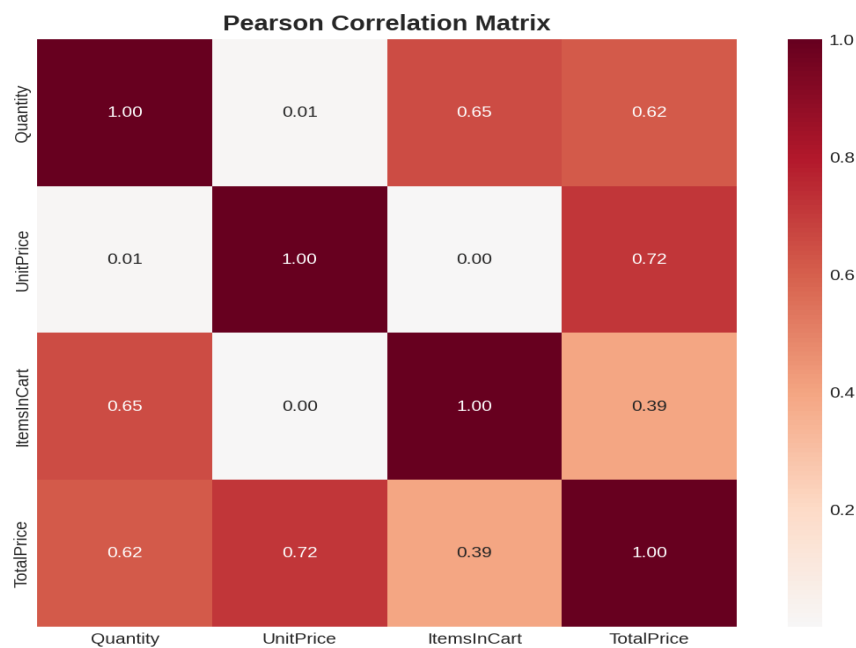
3.3 Outlier Analysis (Noise vs Signal)

Outlier Detection: Boxplots



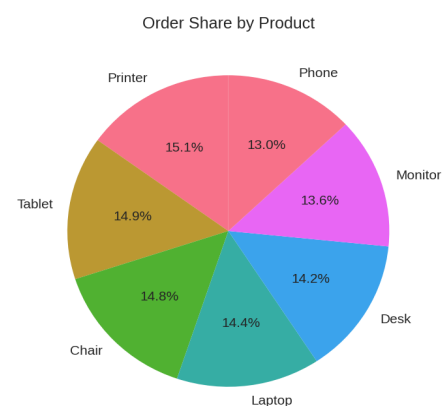
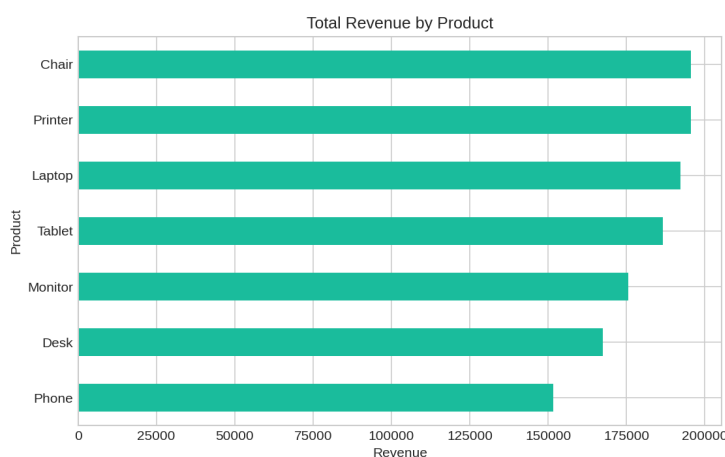
- **IQR method on TotalPrice:** 8 potential outliers above \$3,330 ($Q3 + 1.5 \times IQR$). All are legitimate high-quantity $\times$ high-unit-price combinations (5 units of premium items).
- **Z-score ($|Z| > 3$):** 0 outliers — the distribution is not extreme enough for a normal-based rule.
- **UnitPrice / Quantity / ItemsInCart:** No IQR outliers.
- **Verdict:** The high TotalPrice points are **Signal** (high-value customers), not Noise. Investigate for VIP / bulk-order patterns rather than remove.

3.4 Correlation Structure



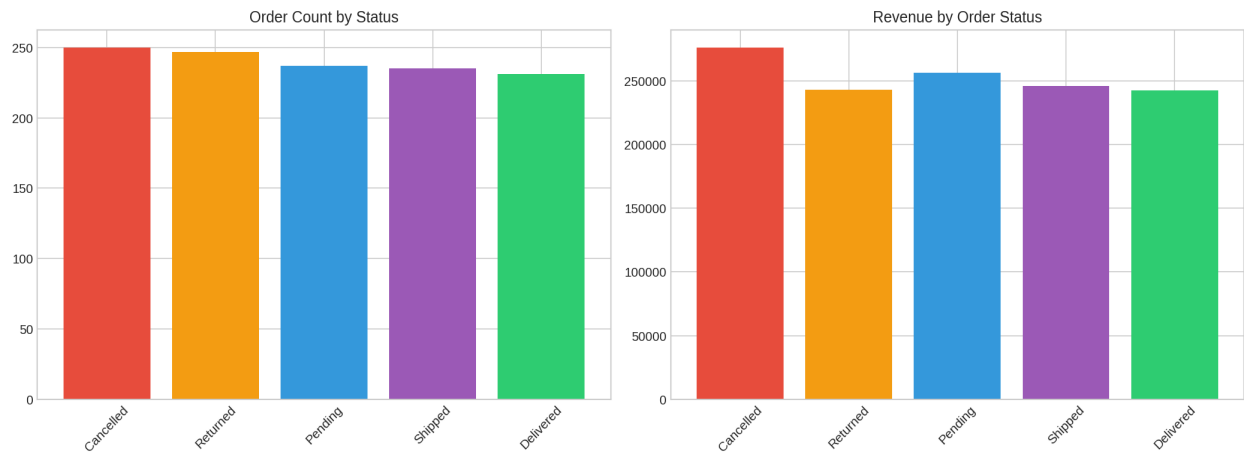
- TotalPrice is strongly driven by UnitPrice ($r = 0.72$) and Quantity ($r = 0.62$).
- ItemsInCart correlates moderately with Quantity ($r = 0.65$) — customers who buy more units tend to have fuller carts.
- UnitPrice is essentially independent of Quantity and ItemsInCart ($r \approx 0$).
- **Golden Rule reminder:** Correlation $\neq$ causation. Higher UnitPrice does not “cause” higher TotalPrice; it is a direct mathematical component.

3.5 Product Performance



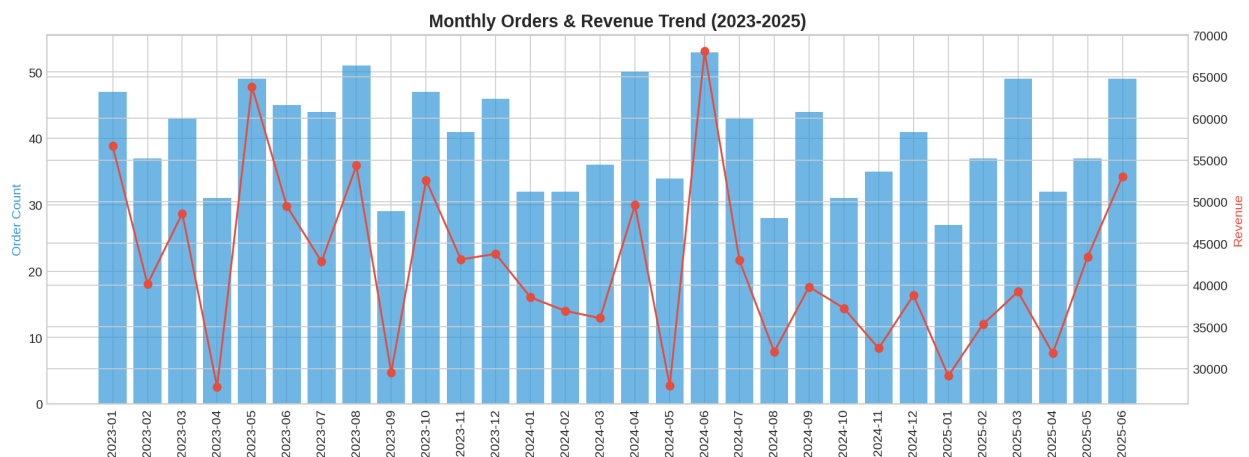
Revenue leaders: Chair (\$195.6k), Printer (\$195.6k), Laptop (\$192.1k). Phone contributes the least revenue despite solid unit volume, suggesting lower average selling prices or smaller basket sizes for that category.

3.6 Order Status – Critical Business Signal



Alarming finding: Cancelled (250) and Returned (247) together represent 41.4% of all orders and \$519.7k of revenue at risk. Delivered orders are only 19.3%. This is a major operational and customer-experience red flag.

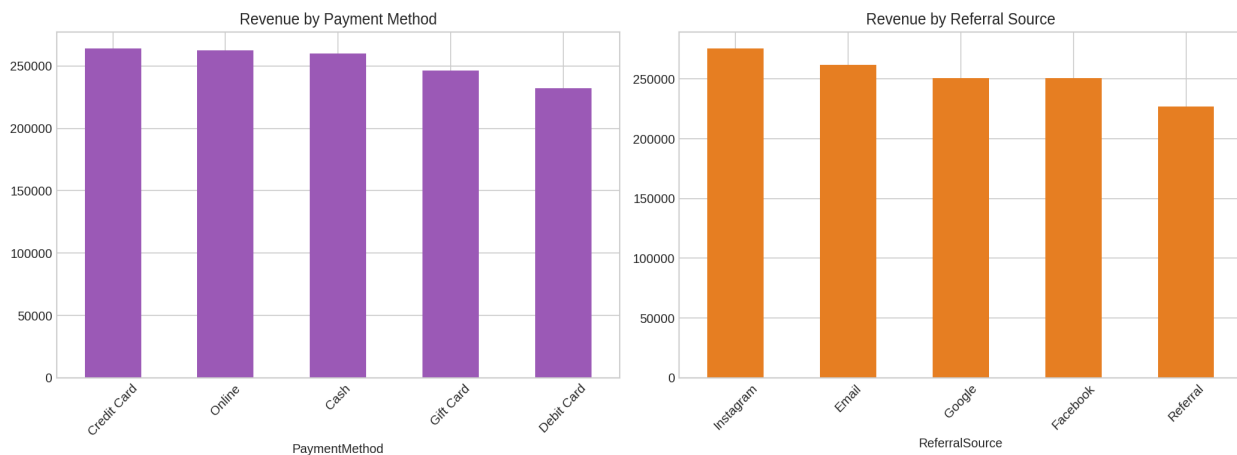
3.7 Temporal Trends



- 2023: 510 orders, \$552.6k revenue (highest full-year performance).
- 2024: 459 orders, \$480.2k (modest decline).
- 2025 (H1 only): 231 orders, \$231.9k — on track for roughly similar or slightly lower annual run-rate.

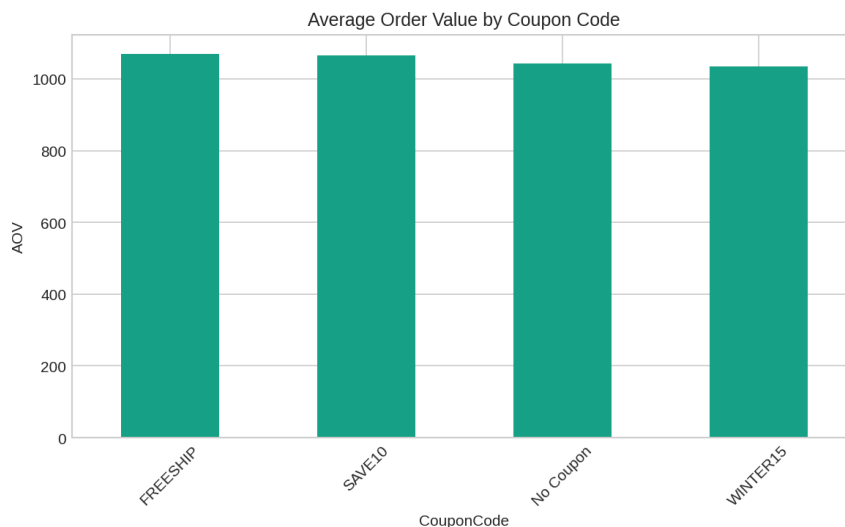
Monthly volume is relatively stable with no strong seasonal spike, though June and August show higher activity in multiple years.

3.8 Payment Method & Referral Source



Payment methods are remarkably balanced (Online leads slightly). Credit Card shows the highest average order value. Instagram is the top referral channel by both volume and revenue, followed by Email and Google.

3.9 Coupon Impact



FREESHIP and SAVE10 coupons are associated with slightly higher AOV than no-coupon or WINTER15. Coupon usage rate is high (74.25%). The lift is modest, suggesting coupons may be used more for conversion than for basket-building.

4. Synthesis – The “So What?” Test

Finding → Business Diagnosis

- 1. Right-skewed TotalPrice** → A small number of high-value orders drive a disproportionate share of revenue. Protect and grow the top-decile customers; do not optimize solely for average order value.
- 2. ~41% Cancelled + Returned** → Nearly half of demand never becomes fulfilled revenue. This is equivalent to losing the revenue of ~497 full orders. Root-cause analysis (product quality, shipping delays, payment failures, inventory) is urgent.
- 3. Instagram + Email dominate acquisition** → Double-down on these channels; test creative and landing-page optimization for Facebook/Referral which lag.
- 4. Chair / Printer / Laptop lead revenue** → Inventory and marketing budget should prioritise these categories.

Phone may need price or promotion review.

5. **Stable but non-growing volume 2023–2025** → Growth is not organic. New acquisition levers or product expansion are required.

5. Recommendations (Actionable Next Steps)

1. **Investigate cancellation & return drivers** – segment by Product, PaymentMethod, ReferralSource and time-to-cancellation. Target a 15-point reduction in combined Cancelled+Returned rate within two quarters.
2. **VIP / high-value customer programme** – identify the top 10% of orders by TotalPrice and design retention offers.
3. **Channel optimisation** – increase Instagram and Email marketing investment; A/B test Facebook creatives.
4. **Coupon strategy review** – test whether FREESHIP can be replaced by threshold free-shipping (e.g., free above \$50) to protect margin.
5. **Next analytical milestone** – build a dashboard for real-time OrderStatus monitoring and a simple predictive model for cancellation risk.

Verdict: The dataset is clean and analytically rich. The dominant signal is operational friction (cancellations & returns) combined with a healthy but concentrated high-value order segment. Addressing fulfilment reliability will unlock the largest near-term revenue upside.

Prepared as part of DecodeLabs Data Analytics Internship – Project 2 (Exploratory Data Analysis).
All calculations reproducible from the supplied Excel dataset using Python (pandas, scipy, matplotlib, seaborn).